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Article

Social Justice in HCI: Current Streams, Considerations, and Ways Forward

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Abstract

The expanding interest in justice-oriented HCI focusing on critical perspectives, structural oppression, and marginalization—often referred to as social justice—is reflected in a growing number of publications over the past few years. Through the continuous growth of social justice in HCI, we argue that now is a good time to provide an overview of the ongoing and current streams of social justice research. We introduce social justice as it has grown in HCI during the last 15 years followed by the most commonly framed theoretical tenets. Secondly, we construct a corpus of 60 HCI articles building on social justice as the main concept. Through our corpus we summarize and present 4 currently ongoing streams of research to further a cohesive, yet fluid understanding of how social justice is shaped and understood within HCI. Describing these interconnected streams also gives us the possibility to frame and describe the current development and move forward as a research community. Based on our study and discussions, we suggest 6 considerations for HCI researchers seeking to work with social justice as a concept, and we emphasize the need for long-term engagement in justice-oriented research to foreground and enable societal change and social good. Through this study, we contribute to the ongoing growth of social justice research in HCI by providing an overview of current streams and ways forward.

RESEARCH HIGHLIGHTS:

- During the last 15 years the concept of social justice has grown in HCI.
- We argue that now is a good time to provide an overview of the current streams and ways forward.
- Through a scoping literature we analyse 60 articles explicitly engaging in social justice.
- We identify four current domain streams to function as a heat map of where social justice is being conducted in HCI.
- We suggest six social justice considerations for HCI researchers and emphasise the need for engagement to foreground social change and social good.
- We contribute to the ongoing growth of social justice research in HCI by providing an overview of current streams and ways forward to foreground a cohesive and fluid understanding of the concept.

Keywords: HCI; Social justice; HCI theory; Concepts and models

1 INTRODUCTION

HCI has been through a transformative shift in recent years, with an increased uptake of principles and perspectives from social justice and parallel concepts such as transformative, restorative, and environmental justice (Corbett and Loukissas, 2019; Dombrowski et al., 2016; Sharifa Sultana et al., 2022; Strohmayer et al., 2019; Sultana et al., 2018). This paradigmatic shift reflects a growing awareness of critical issues surrounding digital technologies as tools of both justice and injustice including e.g. concerns about equity, inclusivity, and fairness (Sharma et al., 2023). The recognition of digital technology as a potential facilitator or mediator of both 'good and bad' has grown to include communities facing large social marginalization and discrimination such as racially discriminated groups, queer communities, indigenous groups, women, and children with experience of physical and sexual abuse, gentrified communities in the United States, and women living under heavy patriarchal structures (Bellini, 2023; Öhlund and Almeida, 2023, 2023; Sharifa Sultana et al., 2022; Strohmayer et al., 2019; Sultana et al., 2018). Social justice opens up for anti-solutionist approaches recognizing how large structural challenges cannot be solved explicitly through technological interventions. HCI literature seeking to frame this concept include both digital interventions, as well as approaches that recognize how large social challenges cannot only be addressed through individual change. For example, (Sharifa Sultana et al., 2022) dissects stigma and shame towards children with sexual abuse experience instead of centering police interventions and (Strohmayer et al., 2019) builds on already existing tools for criminalized sex workers. In doing so larger insights on society and structural challenges are highlighted. To frame such challenges of structural oppression for both micro (community in question) and macro (for the large social challenge) interventions, social justice is often used as an umbrella term dealing with justice and injustice on a larger scale (Bellini et al., 2022; Dombrowski et al., 2016).

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This evolving research landscape—growing in popularity, creates a need for a conjoint understanding of social justice to prevent misaligned interpretations. We extend the arguments by (Bellini et al., 2022) in questioning non-explicit commitments to social justice and the risk to 'run in tension with or counter the goals for co-liberation, equity and redistribution'. The point is not to gatekeep engagement and commitments toward social justice. Instead, the purpose is to build a joint understanding to highlight the opportunities of genuine commitment to social justice in HCI and to prevent misaligned approaches and short-term engagement with the concept. While acknowledging the rapid growth in this area, we contend that a long-term commitment to this strand of research is essential to cultivate a more just future for all. For sure, cultivation and processes of change and inclusion take time. Motivated by this growing area, we present a scoping overview of HCI literature that explicitly frames social justice to highlight the ongoing streams of research in this area.

To structure our paper, we start by presenting how social justice has come to grow in HCI followed by a short theoretical background. We then move to present the method and approach in creating our literature corpus. Moving forward, we present a collection consisting of social justice literature in HCI followed by an analysis resulting in four streams of social justice research in HCI. We conclude, by discussing insights from each stream and suggest 6 considerations for researchers working with social justice as a core concept. Lastly, we question how our overview and streams can serve as a useful tool for the research community.

2 SOCIAL JUSTICE IN HCI

During the last 15 years social justice has grown in HCI. Over these years it has engaged with notions of for example, injustice, oppression, fairness, equity and equality (Bellini et al., 2022; Dombrowski et al., 2016; Fox et al., 2016; Öhlund, 2023; Sharifa Sultana et al., 2022). As a passing concept, social justice is not novel in HCI, however, concrete engagement and foundational analysis was not included until the early 2010's (Antle et al., 2009; Bardzell, 2010; Dimond et al., 2013; Fox et al., 2016). From the very beginning in computing literature, the first article in the ACM digital library—Guide to Computing Literature, explicitly mentioning social justice is 'Computer Security—A Survey' by Peter S. Browne from 1976. Social justice is only mentioned once in the context of privacy as a social concept (Browne, 1976). Most articles between 1975 and 2008 only mention social justice passingly, for example how computer applications may have '*labor-displacement effects and deter social justice*' (Mowshowitz, 1984), how virtual reality in education can stimulate '*fresh perspectives on how to promote social justice*' (Casey et al., 1998) or how '*collaborative inquiry as a method can deploy knowledge towards social justice*' (actually in this article social justice is mentioned 4 times) (Campbell, 2004). Between 1975 and 2008, only 55 articles somehow mentioning social justice were published and it was not until about 2010 that the concept started to gain a more widespread interest. From 2016 and forward, articles on social justice become more frequent, and many authors started to define and theorize more thoroughly, how the concept can be used in for example design, interaction design, UX design and augmented reality research (Ashikoto et al., 2018; Dombrowski et al., 2016; Fox et al., 2016; Rose et al., 2018). Since 2010 social justice has been used in a plethora of variations in HCI, spanning from an umbrella concept discussing any type of larger injustice (Van Wart et al., 2014) to using it as a more specific concept, for a more specific purpose, such as transformative justice (Sharifa Sultana et al., 2022), restorative justice (Sultana et al., 2018) or linguistic justice (Nee et al., 2021). Through the multifaceted use of social justice over the last couple of years, the concept has seen a large increase in interest which has led some researchers to raise concerns over the fast-paced development. One such concern relates to the danger in conducting research, in the name of social justice, without explicitly discussing what the concept really is about (Bellini et al., 2022). Mis- and under-representing social justice efforts can result in injustice through counter-acts of 'liberation, equity and *redistribution according to*' (Bellini et al., 2022). In this paper, by sharing and working for a common understanding of social justice attending to reflexivity and fluidity, we hope to limit the possibilities of running in tension with core tenets of social justice and reduce potential harm to communities involved in such research. Another concern, as the popularity of justice related concept grows, is that of virtue signaling where an individual expresses a certain opinion to gain moral high ground simply based on communicating a good character without any further engagement with justice. This concern is discussed by (de Castro Leal et al., 2021) and echoed by (Bellini et al., 2022) as 'community fetishism' where researchers attain to communities at the margins for rich data seeking career development before social change. We recognize both of these risks of misrepresenting social justice and virtue signaling through community engagement by highlighting the need for a joint common ground for HCI researchers seeking to engage in social justice research. We also build on the recently published work by Chordia et al. (2024) who systematically presents a broad overview of social justice in HCI. They take on a broad scope of justice-related literature in which we extend our scoping review toward social-justice explicit HCI literature.

As this introduction illustrates, there has been a steadily growing interest in social justice oriented HCI over the last 15 years. We also note a shift from the earlier days when the concept was often passingly mentioned, to more recent publications where social justice is an explicit object of study. Accordingly, it is now a good time to map out this growing landscape of research and identify the current research streams for further development of this area. In the following section, we describe the wide theoretical background of social justice.

3 BACKGROUND—THEORETICAL ROOTS OF SOCIAL JUSTICE

Social justice as a theoretical concept relating to justice and fair distribution of social opportunities originates from various historical sources and academic disciplines. Many attribute social justice heritage to philosophical thinkers such as Plato, Kant, and Marx in dealing with various difficult, large, and constant notions such as equality, equity, needs, and just distribution (Jost and Kay, 2010). Moving from social justice as an abstract line of thought to building and framing theory, two researchers are often prominently accredited: John Rawls and David Miller (Jost and Kay, 2010). In the 1971 book, *A theory of Justice, Johan Rawls* foreground one of the largest scholarly works aimed at a theorization of Justice. In the 538-page book, Rawls examines the role of justice in a plethora of paradigms, such as utilitarianism, fairness, morality and one of his most widespread concepts 'the veil of ignorance'. The concept is described through a hypothetical scenario where humans have no knowledge of their position in society and would hence agree on a scheme of equal liberties (Rawls, 1971; Krawczyk, 2010). A later theoretical contribution was written in 2001 by philosopher and professor of political theory, David Miller. In the book 'Principles of Social Justice' (Miller, 1999), Miller expands

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on the justice concept describing it as a set of context-sensitive principles that apply to different forms of human association and identifies three main principles of social justice: *desert, need*, and *equality*. *Desert* frames an individual's rights based on contribution, *need* entails how those in need should get more, and *equality* foregrounds an equal distribution of social benefits. Other influential literature theorizing social justice draws on similar ideas of equality, equity, need, freedom, and human rights such as American philosopher William Frankena who discusses and identifies four main criteria for assessing justice in a society: equality, need, merit, and freedom (Frankena, 1966). Another American philosopher accredited to extending ideas of social justice is Joel Feinberg who emphasizes and defends a liberal and pluralist view on justice, which recognizes the diversity and complexity of human interests and values, as well as the need for a balance between individual and social welfare (Jost and Kay, 2010).

Reflecting on who is accredited to have made substantial efforts to social justice theories in academia common denominators are white, male and western (Jost and Kay, 2010). Despite discussing notions of desert, need and equity, few engage in thoughts on intersectionality—the idea of interconnected experiences of oppression. We extend the scope of previous social justice literature and argue for other important and influential contributions, such as Black feminist thought by Patricia Hill Collins (Collins, 2000), Intersectionality as articulate by Kimberlé Crenshaw (Crenshaw, 2013), Feminist Theory by Bell Hooks (Hooks, 2000) the three-dimensional justice framework by Fraser (2009) and lastly the work on oppressive structures by Iris Marion Young (Young, 2020). These researchers, theorists and activist are often cited and positioned in HCI as core works of justice, feminism, de-colonialism, and social justice (Erete et al., 2021, 2023; Jones, 2023; Ogbonnaya-Ogburu et al., 2020; Strohmayer et al., 2019). The broad usage of multifaceted theories within social justice HCI mirrors the positive growth and development toward more inclusive and critical research.

Working with a concept, such as social justice, it becomes important not only to reflect on the future research we are conducting but also the previous research we are referencing and building on. As the concept of social justice has gained substantial traction from HCI researchers in the last 15 years, and continues to grow, we move to present a scoping overview of HCI literature framing social justice to highlight the current ongoing streams of the area. We do this to build a joint understanding highlighting the opportunities of genuine long-term commitment to the social justice concept.

3.1 Methodology and process

Despite the fairly short history of social justice oriented HCI, over the last 15 years there has been a significant number of papers published that demonstrate a broad interest in this area, and a growing concern for questions dealing with structural challenges of oppression. To capture key literature in this growing field, we prioritized working toward an initial scoping review of literature pertaining explicitly to a social justice concept. Through a scoping review methodology, we had the possibility to present a broad overview of the topic, examine the emerging area and clarify concepts (Tricco et al., 2016). Similar approaches can be seen in articles where researchers in HCI have sought to summarize and highlight concepts such as sustainability (DiSalvo et al., 2010), embodied interaction (Lee-Cultura and Giannakos, 2020), and time and temporality (Wiberg and Stolterman, 2021). Moreover, we build on the recent work by Chordia et al. (2024) who attains to a systematic literature review on articles discussing social justice as an umbrella term aligned with parallel concepts, such as transformative, reproductive, and disability justice. As their work summarizes, for example, the harms and benefits centered in social justice HCI and key considerations on how to think about justice-related questions and concerns we extend their work in a few keyways. Through our smaller scoping review, we highlight HCI literature that articulate explicit attention to the concept of social justice. Furthermore, in difference to their description of common harms, benefits and approaches, followed by tools for equitable processes, we summarize 4 currently ongoing streams to highlight a sort of heatmap where most work is conducted. Lastly, by providing an overview of research which explicitly engages in a social justice concept, we hope to facilitate a larger conversation on what social justice commitments entail and the opportunities of a shared understanding as also highlighted by (Bellini et al., 2022; Chordia et al., 2024).

To start building our literature corpus we conducted a keyword search of ('social justice') in the ACM digital library—Guide to Computing Literature, which is 'the most comprehensive bibliographic database focused exclusively on the field of computing'. Through our first keyword search for ('social justice') proceedings on June 18, 2024, we obtained 150 results (see Figure 1). We downloaded all the results into Zotero and imported the list into a word table for easier overview. In similarity to Chordia et al. (2024) and DiSalvo et al. (2010) we elicited a number of criteria to further refine our corpus also called 'reporting guidelines'. Criteria and reporting guidelines are used as a way of making the methodological approach within a scoping review more transparent (Tricco et al., 2016). Throughout this paper, we use the word *criteria*, more common in HCI when describing iterations of analysis (DiSalvo et al., 2010; Chordia et al., 2024). From the initial 150 keyword results, we created two criteria to ensure that each paper had a strong conceptual connection to social justice beyond the keyword. There were 6 papers which we could not gain access to and for those reasons we excluded them not because of their disconnection to social justice, in which we could not analyze but because of their inaccessibility. This made our total number of proceedings 144. For our following criteria analysis, we excluded any paper which did not (1) Engage with the concept of social justice beyond the keyword or references and (2) Use social justice as a primary concept for the work. For example, is the concept discussed in related research, background, or used as an approach? In for example, (Westbrook, 2015; Payton et al., 2019; Pritchard et al., 2020; Tushar et al., 2020) social justice is not attended to more than in the keyword and although the research topics are highly relevant for social justice research (sex-trafficking equity, inclusion, queer, and feminist theories) for our scoping review we only sought works that explicitly engaged with social justice as a concept. In some cases, we also removed any paper which although mentioning social justice, did not have the space to further engage in the concept due to the format of the proceeding such as short workshop papers or abstracts (Barnard et al., 2021; Wieczorek et al., 2023). However, due to social justice being an abstract and complex concept, we recognize how papers can still be of a social justice nature, and for this purpose, we highlight the broader work by Chordia et al. (2024).

The first author of this paper conducted the initial criteria analysis on the 144 proceedings. By going through each result one by one and searching for social justice in the text, the first author marked down 56 papers as initially passing the first and second criteria and 17 papers marked B for borderline. Seventy-one proceedings were removed based on their disconnection from the criteria. The borderline papers passed the first criterion. But

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did not clearly pass the second. The two authors discussed the 17 borderline papers, and in the end 4 of them were selected for the final corpus consisting of 60 papers.

From these 60 papers, we initially grouped papers sharing a domain and interest. We did this as to get a better overview as to the most common ongoing streams of social justice research in HCI. The corpus was divided into 4 domain and interest streams (1) Social Justice-Oriented Design Research, (2) HCI For Larger Social Justice Initiatives, (3) Technical Communication For Social Justice and (4) Social Justice in Computer Science and Engineering.

Secondly, we developed a scheme of questions similar to DiSalvo et al. (2010) guiding our analysis of each paper. To create the questions, we sought to frame both a cohesive understanding and plurality in the multidimensional usage of the social justice concept in HCI. The questions had to both be straightforward enough to answer and open enough to not discard explorative work. The questions were as follows: (1) How does the paper position and define social justice? (2) Is there a particular type of social injustice? (3) In what ways is the paper engaging in critical perspectives of structural injustice? (4) Is social justice used as a definitive goal or a way of analysis? (5) Who is supposed to take action against the framed social injustice? (6) Are the contributions of the paper aligned with social justice long-term engagement? For each paper, the first author wrote a small summary relating to the questions. Through these summaries, we critically examined and discussed each question working back and forth between our 4 domain streams and collection of papers. Many papers overlapped into another stream, however, to create a more cohesive idea of each domain and interest stream, we decided to place each paper in only one stream. In streams where this was the case, we provide arguments for our choices of placement. Below, we move to present the four most noticeable domain and topical research streams of explicit social justice HCI research.

4 ONGOING SOCIAL JUSTICE STREAMS

Hegemonic domination and structural oppression have engaged many HCI researchers in the last 15 years and social justice has emerged as one way to frame often complex and abstract social challenges. Despite the broad engagement with concepts of oppression and injustice in HCI, social justice is rarely used coherently in existing literature. By summarizing and presenting the ongoing streams of (1) the most common domain and topical interest and (2) how they engage in social justice, the purpose is to provide a cohesive, yet fluid understanding of how social justice is currently shaped and understood throughout HCI. Describing these streams gives us the possibility to frame the constant movement forward and common change in direction for social justice.

4.1 Stream 1: Social justice-oriented design research

Emerging as one of the major and oldest streams of social justice HCI, noted through our corpus are design-oriented studies in HCI consisting of 17/60 papers. Related to this stream we collect literature on UX-design, interaction design, augmented reality and makerspaces for design. Much of the social justice literature centering design builds on the paper and predecessor 'Social Justice-Oriented Interaction Design: Outlining Key Design Strategies and Commitments' by Dombrowski et al., (2016). The article is among the first in HCI making a substantial effort digging deeper into social justice to provide a better contextual understanding for interaction design. As a challenge for the technological paradigm is to explore how social justice specifically aligns with design, the article positions social justice as an orientation consisting of a 'constellation of modes and sensitivities' which includes framing a reflexive commitment to design processes. By treating social justice as an orientation and a 'horizon to work towards' instead of, for example a goal, authors incorporate a more reflexive view of justice as an ongoing process instead of a static utopian future achievable through technological solutions. We recognize the value of treating social justice with reflexivity and openness, as stated by Dombrowski et al. (2016), due to its multifaceted usage. By not establishing a static definition, researchers minimize the risk of one-sided and short-sighted commitments. Playing into this plurality is also the notion of 'wicked problems', characterized by Dombrowski et al. (2016) as societal challenges without clear solutions often in need of political and structural development. Wicked problems as abstract societal challenges without clear definition nor solution mirror the importance of fluid and plural understandings of social justice to better address social complexities. As discussed by Dombrowski (et al., 2016) and in extension Lötter (Lötter, 2011), social justice is seen as an orientation in constant flux, an everchanging flow of multidimensional lived experiences informed by for example history, identity, class, race, gender and ability. As one of the first major works on social justice-oriented design in HCI, many authors have used the Dombrowski paper as their theoretical underpinning, both within the larger design area as well as in other domains such as augmented reality, socio-technical gentrification, food democracy, gun-violence prevention and digital tools to fight stigma (Strohmayer et al., 2017; Ashikoto et al., 2018; Prost et al., 2018; Samuels et al., 2018; Corbett and Loukissas, 2019; Pendse et al., 2021).

In congruence with the work by (Dombrowski et al., 2016), feminist ideas are often attended to in order to frame a larger theoretical scope. Through our corpus and within this stream, we have noted how for example, intersectionality articulated by Kimberlé Crenshaw (Crenshaw, 2013), feminist theory contextualized toward HCI by Shaowen Bardzell (Bardzell, 2010), and the design justice framework by Sasha Costanza Chock (Costanza-Chock, 2020) are literature included to frame a larger scope of interconnected design thinking, feminism and social justice (see: Kannabiran et al., 2018; Tham, 2019; Tseng et al., 2020; Pendse et al., 2021).

Various approaches are taken to interconnect the areas of social justice and UX-design. For example Acharya (2018) frames contextualized UX-design in resource constrained settings as a way to promote social justice while Rose (2016) positions it as a methodology to frame UX toward marginalized communities. For this design-oriented social justice stream, UX-design is considered a way to 'advocate for users' (Walls, 2016) and as having '*maximum impact on marginalized populations through application development*' (quite the statement!). The papers in our corpus discussing UX-design and social justice occasionally positions UX as a component of eventually reaching a state of social justice. When treating social justice as a utopian goal for research it is easy to reproduce solutionist arguments that technological interventions can solve complex social problems. We wish to make authors aware of the risk of such solutionist thinking, and instead note the possibilities of attending to the many frameworks and ideas of social justice already being discussed in HCI such as (Bardzell, 2010; Costanza-Chock, 2020; Dombrowski et al., 2016). In doing this, researchers more constructively articulate their intended

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meaning of social justice and frame a reflexive understanding instead of an un-reachable goal.

As social justice proliferates for designerly work in HCI, we have documented that some papers posit the concept in knowledge and network building in which we acknowledge as a great opportunity to frame long-term engagement to social justice and its development in HCI such as (Antle et al., 2009; Fox et al., 2016, 2017; Oden Choi et al., 2020). Moreover, by seeking to foreground a broadened discourse on social justice in HCI, workshops, short abstracts and panels are a common format to gather researchers and authors. Through for example, workshops, authors can see a value in social justice to encourage easier and more accessible conversations about adjacent topics, such as feminism, anti-colonialism, health technology, and critical design (Kannabiran et al., 2018; Tham, 2019; Gui et al., 2023). Centering research around future suggestions, implications and directions is common for social justice-oriented design research, where much work attends to a long-term commitment in seeking to shape the HCI community toward a more just future for all. This can be seen in for example Prost et al. (2018) where authors discuss how design with local food networks to strengthen food democracy, in Tham (2019) on how to elicit feminist design thinking to better include traditionally marginalized groups, or in Tseng et al. (2020) on how structural social problems need attention when trying to improve the lives of marginalized workers through technology.

For this first domain stream, (1) *social justice-oriented design*, it becomes evident how design is no stranger to the social justice field. Not only is there much empirical literature, but authors have also moved to develop theories specifically aligned with design and social justice such as in (Dombrowski et al., 2016). Therefore, we highlight design as one of the larger HCI domains engaging in social justice work and how a majority of this work positions long-term engagement—which we argue is of significant importance for the HCI field and its future development.

4.2 Stream 2: HCI for larger social justice initiatives

Moving from the manyfold usage of social justice in design, we are now taking a turn to HCI research which in different ways uses social justice as a lens to explicitly frame complex social challenges such as but not limited to, poverty, trans rights, class, fat positivity, racism, older adults rights, sex-worker rights and gentrification (Edenfield et al., 2021; Corbett and Loukissas, 2019; Jonas et al., 2024; Lan Fang, 2022; Öhlund, 2021; de Santana et al., 2024; Strohmayer et al., 2017). With the surge of HCI research seeking to address historically rooted challenges and injustice, social justice has grown as an approach within this critical paradigm which stands for 16/60 papers from our corpus. For this stream, we include research which first and foremost share the common interest of structural change framed through technology (see Edenfield et al., 2021; Corbett and Loukissas, 2019; Öhlund, 2023; Strohmayer et al., 2019). This is a common horizon amongst all our streams as it is one of the main pillars of social justice. What sets this stream apart from for example, design is that the research included often report on large structural and historical issues first and secondly discuss how technology and social computing can attain to these challenges. We note the difference between stream (1), where often the design and development of an artifact are the main goal at the beginning of each project. For this stream, outcomes and implication for structural change are often discovered throughout the project together with participants looking for structural change such as in (Chen & Bergholm, 2020; Öhlund, 2023; Strohmayer et al., 2017).

By framing for example gentrification, the phenomenon of forced displacement of urban communities by wealthier property investors, HCI have the potential to address the complex challenges through a social justice analysis (Corbett and Loukissas, 2019). At first glance one might consider HCI-scholarship and gentrified communities as two worlds apart, however early HCI work established how the world of technology does not stand apart from the physical world (Dourish, 2006; Harrison et al., 2007). By recognizing how the technological realm does not act in isolation from the lives of humans and in extension the oppression of human lives, social justice provides guidance in analyzing the larger frameworks of injustice. Furthermore, in Strohmayer et al. (2017), authors frame sex work and aid organizations as domains where intricate relationships of legal frameworks, social stigma and discrimination dominate. Through a social justice lens, it becomes possible to address this complex field and highlight the relevance of technological interventions. This is also explored in the context of highly exposed social media moderators in Steiger et al. (2021).

Moreover, we note that some authors incorporate discussions on democracy as part of a social justice concept. In for example Kariotis and Mir (2020) authors explore the dangers of 'black boxing' algorithmic governance and frames participatory methodologies and a social justice lens as a part of bettering transparency and democracy. Similarly, Johnson and Crivellaro (2021) bring together a panel of 'distinct citizens' to create discussions on social justice HCI research proposals to foster civic engagement. This type of participatory approach is common within social justice for HCI to better include marginalized communities as a way to bring in voices who have structurally been left out from society. These voices come from a multitude of communities and are often positioned as essential to include when researching social issues which affects them (Rabaan, 2021; Lan Fang, 2022; Öhlund, 2023; Jonas et al., 2024). One such group in which many papers of this stream attends, is the older adults/senior community. When discussing technology use during COVID-19 among older adults, Öhlund (2023) argues that a social justice orientation can help researchers stay away from ageist arguments on how seniors always need to adapt to technology to live better lives. Similar arguments are presented by Sixsmith (2022) which discusses a social justice rights-based approach to AgeTech development, and by Lan Fang (2022) which discusses social justice disparities which can increase vulnerability of older adults.

In this stream, we also find papers which takes a larger grip on the HCI social justice community in for example providing one of the first larger literature reviews on the topic foregrounding a broader cohesive scope of the concept (Chordia et al., 2024). We also find papers discussing common pitfalls such as shortsighted goals and virtue signaling of the social justice concept (Bellini et al., 2022). What moreover unifies this stream is that the larger take on society and structural oppression through social justice often comes with realizations and conclusions on how big and complex challenges need equally complex attention and intervention. In for example de Castro Leal et al. (2021), authors describe how the social justice turn in HCI often foregrounds a complex activist stance, discussing possible ways forwards to 'contribute to a more just world'. In addition Klassen (2023), frames her future social justice research project to attune 'disenfranchised communities' while Marks and Stanfill (2023) seeks to provide initial steps toward a multidisciplinary field called 'social justice informatics'. This field would bring in a better composition of social justice toward all areas on information and technology as they build on the ideas by philosopher Iris Marion Young in that

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'as long as an inhabitant of this world suffers institutional domination and oppression, they are a legitimate subject of justice'.

In summary of this second stream (2) HCI for Larger Social Justice Initiatives, we have discussed work that first and foremost bring in seemingly un-technologically related issues such as gentrification, sex work, racism, sexism, ageism and many more topics relating to structural oppression. This stream brings in arguments of participatory practices, democratic rights and long-term goals for a better world highlighting the importance of socio-political development together with socio-technical means.

4.3 Stream 3: Technical communication for social justice

This third stream, which makes up for 17/60 papers of our corpus consist of literature which mostly attends to the technical professional communication (TPC) domain. For this stream we went back and forth between papers in which many could fit in other categories. One such example is Ganguly (2021) in which we first sought to put in the second stream due to its scope on reproductive justice. However, after carefully reading the paper, contributions were centered toward TPC scholars in particular and not HCI in general. Hence, we have sorted the paper in this third stream. TPC attending to social justice often foreground the concept through engagement in reflexive practices of domination and injustice (Bay & Ruiz, 2020; Swacha, 2022). These more open practices can for example be obtained through a substantial inclusion of marginalized and under-represented communities which reveals a common approach for TPC and social justice work (Bay & Ruiz, 2020; Rose, 2016).

We quickly note that most papers of this stream base their understanding of social justice from the book 'Technical Communication After the Social Justice Turn' by Walton, Moore and Jones (Walton et al., 2019). The book discusses how social justice has grown in TPC research and positions conceptual understanding and the 4R's (recognize, reveal, reject, replace) framework to recognize how TPC can be part of but also work against oppression. In referencing this social justice framework of the 4r's we note a multitude of papers discussing for example activism on twitter for reproductive and institutional justice (Ganguly, 2021; Jones, 2023; Jones, 2021). We note the paper 'Mapping Out the Core Constructs of Social Justice' (Ojedele-Adejumo, 2023a) which explicitly seeks to build on Walton et al. (2019) 4R's model to conceptualize justice for the TPC field. The same author critiques Waltons and Jones in arguing that emancipation as a form of social justice is rarely included in relevant literature building on their framework (Ojedele-Adejumo, 2023a).

In discussing TPC pedagogy, or 'TPC for the classroom', the Walton and Jones book is used to frame conversations on social justice, marginalization, disempowered communities and critical reflection (Flanagan, 2020; Bay & Ruiz, 2020; Marks and Stanfill, 2023; Swacha, 2022). In for example Lane (2018) authors draw attention to social justice as a way to address the complexities of 'writing for the public'. By using design thinking to draw attention to 'unconventional topics' such as criminal justice, they discuss TPC student courses with the potential to become socially engaged. Additionally, to the TPC contextualized theory and framework by Walton et al. (2019) we note how other social justice related theories are often incorporated in this literature such as systemic oppression by Iris Marion Young (Jones, 2023), Martha Nussbaum on human Integrity (Ojedele-Adejumo, 2023a) and social justice-oriented interaction design by Dombrowski (Ganguly, 2021). Moreover, Rose and Walton (2015) argue that using a lens of posthumanism can contribute to the social justice turn in TPC, while Smith (2023) seeks to bring in a larger scope of feminist methodologies to guide TPC scholars 'towards justice'.

For this third stream we note that in TPC, it is not always clear what authors mean when engaging in a social justice concept. In for example, Stevens (2022) social justice is discussed as issues, perspectives, and something than TPC can enact. Similarly, in Tang (2020), authors discuss social justice both as problems and as a concept which can be advocated for when designing automatic writing technologies. This type of scattered usage can also be seen in Flanagan (2020) where social justice is described as a methodology, something to be advanced and a type of issue. By drawing on experiences of women exploring user centered approaches for home pregnancy tests, social justice helps to shape research and outcomes in (Opel, 2014). Authors explore how social justice can be promoted through their research and despite not providing an explicit definition or deepened argument on what the promotion of social justice means, authors discuss the potential power of the health and medical industry to 'affect a just means of distribution of social goods to all user, including users who are not normatively recognized' (Opel, 2014). This connection to distribution as a means of justice is common and dates back to many original tenets of social justice as seen in (Frankena, 1966; Miller, 1999).

Through our literature corpus we understand that social justice is on the rise in TPC literature as it makes up for 17/60 papers, with theoretical development in the form of books and papers, more published material on the topic and building on related social justice literature. This development seems to follow along the same pattern as in stream one where social justice has seen a larger theoretical interest only a few years earlier than in TPC. We see this as a positive development and note the many opportunities of creating a joint understanding of how social justice can be used for long-term commitments toward equity, fairness and justice in both HCI and TPC.

4.4 Stream 4: Social justice in computer science and engineering

For this fourth stream we bring together literature which lies at the intersection of HCI, computer science and Engineering and makes up for 10/60 papers. Most literature attends to educational approaches in computer science to engage students with social issues and explore how the computing field has a growing responsibility toward complex social challenges. For example, Van Wart et al. (2014) describes an after-school course focused on computer science learning with real world social justice challenges. Comparably, in the paper social justice is a highly mentioned concept used in discussing engineering students' engagement with social justice issues. Jiménez et al. (2019) centers Gutstein's (2012) notion of *a social justice pedagogy and* Fraser (2009) idea of three-dimensional justice. These frameworks assist students to critically reflect on social issues in their community which can according to the authors be a challenging connection to make.

Ferreira and Vardi (2021) highlights the importance of educating CS students on 'the large-scale social implications of their actions' and bring in discussions on the ambiguity of the social justice concept however concretizing that working toward any justice should include justice for *all*. Parallel arguments of critical education in cs can be seen in (Mayhew and Patitsas, 2023) seeking to 'enact social justice' in the classroom, in (Padiyath et al., 2024) when discussing critical anti-oppressive approaches to cs education and in (Fonteles et al., 2024) in using a 'social justice-based framework' to explore the experiences of underserved students. Much like many other computing areas CS and CS education have come a long way in the inclusion and usage of various social

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justice ideas. As described above, many authors and educators seem to have realized the weight that critical CS education can have in today's society not only on the students themselves but also the potential outcomes of future CS developments. To critically examine the initiatives of justice-centered approaches to computer science Lin (2022) seeks to bring in larger questions of socio-political values of CS, resistance of dominant computing culture and framing CS learning as a source for social justice. All in all, authors emphasize a stronger political vision for the reshaping of cs education to foreground a better and more just future for all.

Most papers have a clear idea of whom is supposed to enact or take initiative on social justice within this stream. Either it is the teachers such as in Mayhew and Patitsas (2023) and Padiyath et al. (2024) or for the students to learn and in their later career use such as in Ferreira and Vardi (2021) or Lin (2022). Even though not always clearly stating exactly what is meant by a social justice approach, framework or issue, it becomes less abstract when bringing in clear takeaways and implications. For example, in Zhu et al. (2024) authors present the Robots for Social Justice framework which is meant to give better tools for engineers to 'enhance human capabilities'. Despite never bringing in any specific theoretical framework for social justice or providing any larger defining frames, by engaging in Martha Nussbaum's 10 central human capabilities we are given a framework of guidance. Moreover, through the Engineering for social justice framework, 5 recommendations for future development and a set of questions for better guidance toward equitable futures it becomes easier to grasp what social justice entails. In a Keynote from 2017, Evans (2017) critiques CS research as male dominated and in particular badly adapted to for example female participation in conferences, academic buildings and the challenges of the academic career for women. In his keynote social justice should not only be adapted within the research field and its curriculum but something that should be visible throughout all layers of academia. One of the few papers from our corpus which discusses social justice in connection to the more 'conventional' social justice-philosophers such as John Rawls and Emanuel Kant is (McMenemy, 2021). The paper is published within the library and information ethics, (which we were not sure in which stream this paper belonged) and goes into depth on the concept of communitarianism, liberalism and neutrality which are central concepts in many older academic social justice literature (John, 1971; Jost and Kay, 2010; Miller, 1999).

This fourth and final stream of social justice literature in HCI, engages in a plethora of computer science and engineering research. Like the other streams, this fourth stream is both growing in publications and engagement. A majority of the literature in this stream attends to teachers and students who are to become computer scientists. We note this as an important focus point in how this facilitates long-term commitment of social justice toward future scholars.

5 TIMELINE AND DISCUSSION

In this paper we have introduced how social justice has grown in HCI over the last 15 years, to provide a cohesive, yet fluid understanding of how the concept is shaped and understood in HCI. Through a literature corpus consisting of 60 social justice papers, and the identification of 4 current domain streams of social justice research, we have presented an overview of how the field is progressing forward. Motivated by the methodological approach taken by DiSalvo et al. (2010) to 'map out' where our field of research is going, we have centered our research aim along their formulation to 'map out the approaches being taken and the intellectual commitments that underlie the area, to allow for community discussion about where the field should go'. We also build on the recently published paper by (Chordia et al., 2024) who attains to a systematic literature review on social justice as an umbrella term aligned with parallel concepts such as transformative, reproductive and disability- justice. As we provide a narrower scoping analysis of social justice literature in HCI we see great opportunity for both papers to be used as foundation for community discussion and development. Before we move to discuss each stream and come back to our guiding questions of analysis, we present and discuss a timeline of each stream for easier access and overview of the corpus. We then proceed to summarize our insights from each stream and finally suggest six considerations based on our guiding research questions.

5.1 Timeline

To show how the streams have evolved through the years starting in 2009 when the first paper from our corpus was published until 2024, we created a diagram which presents the number of stream publications each year (See image 1.). The initial year visually represented through this diagram is 2009, in which our corpus begins. However, it is not until about 2016 that a steady stream of publications in all streams started. Through this diagram we can also note how the first stream of social justice-oriented design, with 17/60 papers, follows along most of the years starting with the first ever title social justice publication in 2009. Design have been a growing social justice domain as shown through our corpus although in the later years it seems that design has slowly taken halt. We argue that the halt in social justice HCI design papers is probably not because of a lack of interest but because many design researchers have begun taking on a larger scope of social justice in which we have then sorted the papers in stream 2. The second stream including research which use social justice as a concept to address large scale social challenges starts in 2017, grows in 2019 and continues until 2024 making up for 17/60 papers. This growth highlights how HCI research has in recent years started to show interest in larger nuances of critical thinking and socio-political change. The third stream of technical communication for social justice is also a growing stream with publications in most years from 2014 also being one of the biggest streams in our corpus with 16/60 papers. The fourth and final stream represents literature from our corpus which interlinks HCI, computer science and engineering mostly attending to teaching and pedagogy. As this streams gradually grows with the most publications in 2024, we suspect more research is to come with the high rise in interest of artificial intelligence and its social effects.

5.2 Moving with the streams(s)

From this point, we bring in insights from all the four streams to provide a conclusive discussion on how to use this paper moving forward. As we set out to write this paper with the purpose of providing a cohesive, yet fluid understanding of how social justice is currently shaped and understood throughout HCI we furthermore come back to our guiding research questions. In doing so, we suggest six considerations for researchers wanting to work with social justice which we hope can gather the community toward a more coherent and reflective usage of the concept.

In the first stream of social justice-oriented design-research we found that much work builds on the paper by Dombrowski et al., (2016) and extends their idea of social justice as an orientation and a horizon. In doing this, many authors build new ideas on

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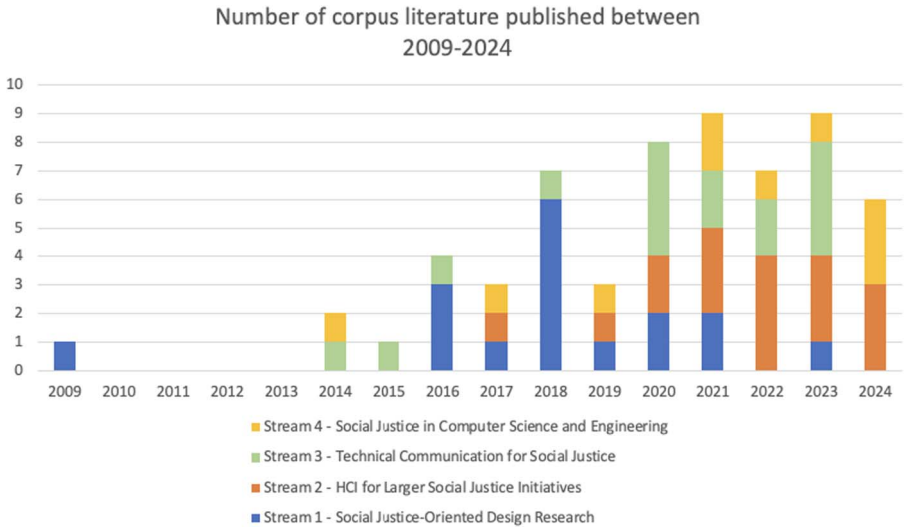


Image 1. Diagram over corpus literature published between 2009–2024

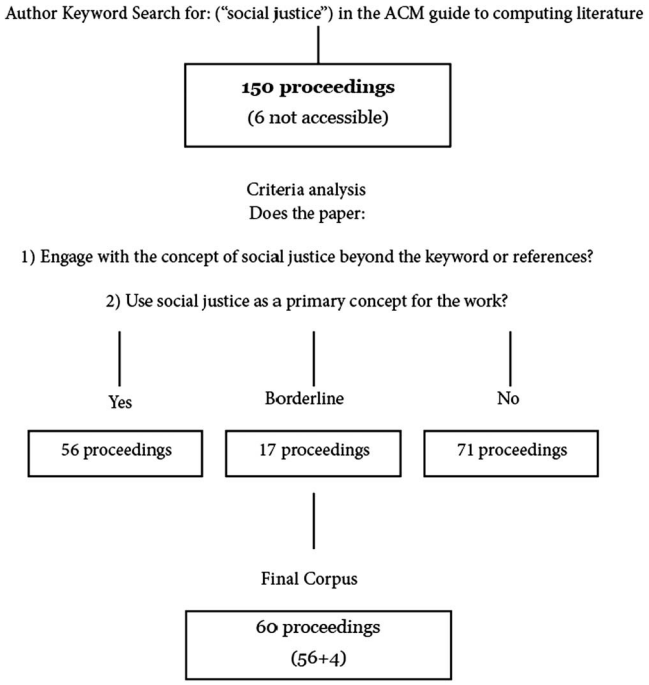


Figure 1. Overview of the selection process

already existing parameters making it easier for future research to follow and to further a cohesion of social justice usage. Besides building on Dombrowski et al. (2016), we see how many authors bring in for example theories on feminism, intersectionality, and design justice which can also act as an adequate way of framing social justice (Kannabiran et al., 2018; Tham, 2019; Tseng et al., 2020; Pendse et al., 2021). We note how these critical theories are more commonly foregrounded in HCI rather than in other social justice literature such as the work by David Miller and John Rawls. There is no one particular topic that is addressed in the design stream and instead much literature touches on a variety of connected areas such as UX and interaction design in correlation with important critical analysis of equity, participation, stigma and marginalization (Acharya, 2018; Corbett and Loukissas, 2019; Tseng et al., 2020; Pendse et al., 2021). We also note how shorter formats such as abstracts, workshops, short papers and keynotes is a commonly used format used for broader discussions.

In our second stream of HCI literature in which directly addresses large complex social challenges we found an array of fields interlinking the socio-technical with the socio-political. Papers within this stream often make the effort to concretely explain and present the complexity of challenges such as gendered violence, ageism, and disenfranchised communities (Strohmayer et al., 2017; Klassen, 2023; Öhlund, 2023). We argue that in this stream we most easily find and connect contributions to long-term engagement of social justice. In discussing democratic rights and problematizing how sociotechnical solutions are not viable as the end goal of research, literature from the second stream most often realize how nuances of history, culture and society effect participants and outcomes. Moreover authors often highlight how contributions can be used and shaped toward a more equitable future (Johnson and Crivellaro, 2021; Kariotis & Mir, 2020; Strohmayer et al., 2017).

The third stream centers the field of technical (professional) communication and show how the area in much likeness to design, have opened up for a larger theoretical underpinning and more publications in a phase of development as seen in (Ganguly, 2021; Jones, 2023; Jones, 2021; Ojedele-Adejumo, 2023a, 2023b). Many scholars build on the book by Walton et al. (2019) in discussing social justice through the concept of 4R's (recognize, reveal, reject, replace). There has also been some community critique in seeking to better strengthen the framework for future usage (Ganguly, 2021; Jones, 2021; Ojedele-Adejumo, 2023b, 2023a). Despite not always using social justice in a coherent way, sometimes describing it as methods, issues, goals, frameworks, etc. implications or takeaways are often directed at either TPC researchers or TPC teachers, two concrete groups (Flanagan, 2020; Stevens, 2022; Tang, 2020). Overall, we see the development of theory, community critique and increasing rate of publication of social justice literature in TPC as highly positive and hope that the movement toward a larger scope of social justice continues.

Our fourth and final stream attends to literature of computer science and engineering. Here, most of the literature focuses on how computer science education can and should be shaped with a larger attention to justice and equity. In contexts of after-school programs, critical reflections of social implications, anti-oppressive approaches to teaching, critical CS education is

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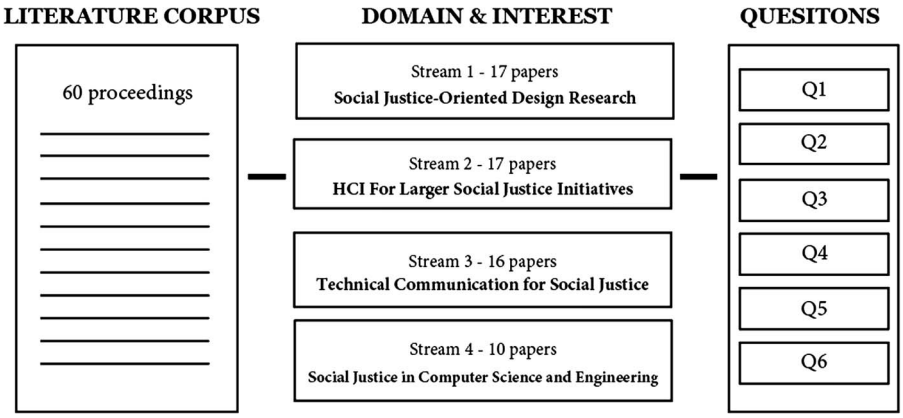


Figure 2. Process of corpus analysis

growing (Van Wart et al., 2014; Ferreira and Vardi, 2021; Lin, 2022; Mayhew and Patitsas, 2023; Fonteles et al., 2024; Padiyath et al., 2024). We welcome a higher sense of critical urgency in CS education considering the exponential growth of artificial intelligence and its power in society. It is also noticeable in this stream how social justice is rarely articulated or defined in a coherent way however through theories like the capability framework by Nussbaum (Zhu et al., 2024) and Frasers three-dimensional justice framework (Jiménez et al., 2019) readers are guided theoretically.

Based on our 4 domain streams of social justice HCI research and our above insights, we move to contextualize and provide initial answers to our guiding research questions.

5.3 Social justice considerations for a coherent future

Coming back to our guiding research questions: (1) How does the paper position and define social justice? (2) Is there a particular type of social injustice? (3) In what ways is the paper engaging in critical perspectives of structural injustice? (4) Is social justice used as a definitive goal or a way of analysis? (5) Who is supposed to act against the framed social injustice? and (6) Are the contributions of the paper aligned with social justice long-term engagement? We have some initial answers. To answer the first question on how the literature within our corpus and throughout each stream defines and positions social justice the most truthful answer would be—in many different ways. Through frameworks, (Ganguly, 2021; Flanagan, 2020; Kariotis & Mir, 2020; Zhu et al., 2024), orientations (Dombrowski et al., 2016; Pendse et al., 2021; Prost et al., 2018; Samuels et al., 2018), adjacent critical theories (Chen & Bergholm, 2020; Jones, 2023; de Castro Leal et al., 2021; Prost et al., 2018; Strohmayer et al., 2017; Tham, 2019), and issues (Jiménez et al., 2019; Tang, 2020; Stevens, 2022) we note the big difference in usage and the lack of cohesion. As we work for a common understanding of social justice attending to reflexivity and fluidity, we do not see the need for only one way of discussing social justice, however, we do wish to make future authors reflect on the following: *Consider articulating how and what social justice means for your particular research context*. By doing this we argue that social justice can become a more accessible concept for researchers to build on and in the end further long-term engagement through cohesion.

The second question, (RQ 2. *Is there a particular type of social injustice?*) is similarly to the first one, not easily answered with a single word. However, if we look at the particular streams, we can note how in stream 1,2 and 3, there is a higher level of including groups which have experienced marginalzation , such as black and people of color, sex-workers, older adults, women and children at risk of violence and abuse, and transgender individuals (Edenfield et al., 2021; Jones, 2021; Klassen, 2023; Lan Fang, 2022; Mudliar, 2020; Öhlund, 2023; Strohmayer et al., 2017). In the fourth stream of corpus literature which includes the CS and Engineering fields, most of the context deals with educational approaches to CS and social justice analyzing how the CS field are in need of more socially critical discussions (Van Wart et al., 2014; Jiménez et al., 2019; Ferreira and Vardi, 2021; Fonteles et al., 2024). From our second guiding research question we suggest authors to: *Consider whom are at the end of the social injustice you are attending to in your research*. As we have already noted through our streams, this is not uncommon in HCI but on the other hand a positive commonality in which we hope will continue in the research community to a bigger degree.

Our third guiding research question asks in what way each paper claiming to work with social justice actually engages in larger questions of structural critical analysis (RQ 3. *In what ways is the paper engaging in critical perspectives of structural injustice?*). Much of the literature in our corpus often foreground a plethora of intersectional directions instead of the more common status quo justice literature by for example John Rawls and David Miller as suggested by (Jost and Kay, 2010). We are convinced that when engaging in theories of a broader scope of inclusion, through for example, Kimberlé Crenshaws ideas of intersectionality (Tham, 2019), Iris Marion Youngs notion on systemic oppression (Chen & Bergholm, 2020; Jones, 2023; Tacheva et al., 2022), the extended thoughts on feminism in HCI by Shaowen Bardzell (de Castro Leal et al., 2021) or the multi-dimensional justice framework by Fraser (Jiménez et al., 2019; Strohmayer et al., 2019) authors are actively working to shape the field for better inclusion of critical analysis. We note how critical positioning is important as to limit using social justice as a signpost for virtue signaling and suggest authors to: *Consider what theory or framework you are using to position social justice as a large socio-critical concept*. By framing this consideration, we hope to make authors aware of the often-sociopolitical nature of social (in)justice and how this can run in tension with for example suggestions of heightened consumption and production.

As we extend the arguments by (Bellini et al., 2022) in questioning non-explicit commitments to social justice and the risk to 'run in tension with or counter the goals for co-liberation, equity and redistribution' we used the fourth guiding research question (RQ 4. *Is social justice used as a definitive goal or a way of analysis?*) to openly look for authors considering social justice as

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the end goal of research. While this may not be directly negative, we caution authors framing their socio-technical interventions as socio-political solutions. Many of the social challenges and issues which are explored in social justice HCI research such as racism, sexism, ageism and homophobia are too big and complex to 'solve'. Claiming that certain technological interventions (on their own) might entail a large-scale social change should be carefully considered by researchers as solutionist rhetoric. Social justice is a broad concept, and an area utilized in many ways, however, by reproducing the notion of technology as the key to socio-political change, researchers are actively going against many tenets of social justice as for example anti-capitalistic. Social justice HCI has the potential to frame large-scale socio-political challenges; however, we should tread lightly when implying capitalistic production of digital artifacts as solutions. We suggest that authors: *Consider how you position the end goal of your research, are you implying heightened consumption and production of products?* Despite our arguments of careful considerations toward solutionist framings, we note how much of our corpus literature applies social justice in a reflexive and constructive way, often building on earlier HCI research. This is especially the case for design where social justice has been developed vicariously to better suit the domain and its various approaches, most know through the paper by (Dombrowski et al., 2016).

Our fifth guiding research question *(RQ.5 Who is supposed to act against the framed social injustice?)* asks who is supposed to take action against and change current challenges and issues. When discussing for example democratic change (Johnson and Crivellaro, 2021; Kariotis & Mir, 2020), educational change (K. Flanagan, 2020; Lin, 2022; Mayhew and Patitsas, 2023), or community change (Corbett and Loukissas, 2019; Bellini et al., 2022; Chordia, 2022), who is thought to make the effort? This is an important part of social justice work in order to prevent suggestions of interpersonal change where individuals are supposed to adapt and change their lives when seeking better futures instead of focusing efforts on the bigger context which is inevitably what causes injustice. We therefore suggest the following: *Consider clearly articulating who is supposed to use, buy, learn, adapt, and in the end change to each research outcome in order to prevent misaligned responsibility which can elicit injustice.* We argue that this consideration can better assist researchers to reflect on where responsibility and power of change lies.

Our last and final guiding research question *(RQ.6 Are the contributions of the paper aligned with social justice long-term engagement?)* calls for long-term commitment which is inevitably an important part of social justice HCI research which is also noted through our corpus (Antle et al., 2009; Bellini et al., 2022; Chordia, 2022; Fox et al., 2016; Oden Choi et al., 2020; Öhlund, 2023; Tseng et al., 2020). Such large social challenges framed in much of our corpus cannot be addressed only through small research efforts and instead demands long-standing attention throughout many years and initiatives. To highlight the importance of framing social justice research for long-term commitment, we finally suggest that authors: *Considers how your work can be continuously built on in order to frame long-term commitment to social justice.* We recognize the difficulty of this since many research areas shift throughout the years because of trends in academia. However, if we continue to shape research contributions to foreground more accessible continuation, we build stronger foundations for long-term engagement in social justice HCI.

Moving forward, we ask: *how can our overview and the identified streams serve as a useful tool for the research community?* Firstly, we mean that our streams pose as an initial insight into where social justice has grown in HCI and in which direction its moving. We argue that it is too early to make any definitive statements on where social justice should or could move, however, we hope that our paper can foreground more accessible community discussions and future research. Moreover, we suggest that our paper does not only provide an overview but has the potential to work as a 'heatmap' of the field showing where the most and least research is conducted. An understanding of this distribution can help to identify research gaps, emerging strands or topics, or where mainstream research in this area has been and is being conducted. We can for example, already note how the design domain is seemingly one of the largest area where social justice is established and developed as opposed to the Computer science and engineering domain where research is limited and still growing. This is helpful in the planning of further studies, and it provides a map where we as a research community can start to fill out the blank spots. Another opportunity is related to an increase in our ability as a research community to critically examine new research in this area. To which domain is research a contribution? What do we already know? How does a new study challenge what we thought we knew? In short, strengthening the collective research discourse and long-term engagement. Here, we suggest that this can be done through the following six considerations:

- *Consider articulating how and what social justice means for your particular research context.*
- *Consider whom are at the end of the social injustice you are attending to in your research.*
- *Consider what theory or framework you are using to position social justice as a large socially critical concept*
- *Consider how you position the end goal of your research, are you implying heightened consumption and production of products?*
- *Considers how your work can be continuously built on in order to frame long-term commitment to social justice*

Finally, most research areas develop slowly, over time. Accordingly, a systematic approach to social justice research need to account for this slow-paced temporality and apply a long-term research agenda. This foregrounds us to understand how society, as well as research in HCI, is slowly changing, moving, reproducing—over time, and how to critically study and center societal impact. Social justice in itself is also complex, and to bring about change in this area is a path pawned with challenges. So again, there is no easy way forward, no quick fixes, ready-made solutions or short cuts. It will take time, and accordingly it needs a long-term commitment. While an overview can be thought of as a tool to identify research gaps and enable us to quickly move from one study to the next, our perspective is quite different. We think that societal impact in this area might not be just about projects, small-scale interventions, or made possible through various design experimentations being conducted, but rather that real change happens on an aggregated systemic level. That is, through how individual and small steps taken, together, collectively, steadily, and continuously help us to move forward together—and toward a more just future—for all.

6 CONCLUSION

As justice-oriented research has grown in HCI, we have in this paper presented a scoping overview of four current domain and topical streams of social justice research. We started off by introducing social justice and its growth in HCI during the last 15 years as well as outlining theoretical tenets. Based on 60 articles

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collected from the ACM digital library—Guide to Computing Literature, we constructed a corpus of literature building on social justice as one of the main concepts. Describing these interconnected streams gave us the possibility to frame and discuss the current movement of social justice in HCI highlighting community discussion and long-term engagement. We moreover suggested 6 considerations for HCI researchers to reflect on when working with social justice as a concept. By doing this we aimed at providing a cohesive, yet fluid understanding of how social justice is shaped and understood throughout HCI. Through our work and this study, we contribute to the ongoing growth of social justice research in HCI and finally emphasize the need for long-term engagements. This is necessary not only for social justice as a research topic in HCI, but as an approach to strive for and enable societal change in the world—for all .

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Data availability

The data underlying this article are available in the article and in its online supplementary material.

Conflicts of interest

There are no conflicts of interest to declare.

References

Acharya, K. R. (2018). Usability for user empowerment: Promoting social justice and human rights through localized UX Design. In *Proceedings of the 36th ACM International Conference on the Design of Communication*, 1–7. https://doi.org/10.1145/3233756.3233960

Antle, A. N., Corness, G., & Droumeva, M. (2009). Springboard: Exploring embodiment, balance and social justice. In *CHI '09 Extended Abstracts on Human Factors in Computing Systems* (pp. 3961–3966). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/1520340.1520601

Ashikoto, L. I., Ajibola, D., & Virmasalo, V. (2018). A room for social justice: Affective and interactive augmented reality exploration. In *Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities*, Article 58 (pp. 1–4). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3283458.3283505

Bardzell, S. (2010). Feminist HCI: Taking stock and outlining an agenda for design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 1301–1310). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/1753326.1753521

Barnard, E., Russo, A., & Arriaga, G. (2021). Purposeful plans for computer science diversity, equity, and inclusion. In *Proceedings of the 52nd ACM Technical Symposium on Computer Science Education* (p. 1362). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3408877.3439529

Bay, J., & Ruiz, V. (2020). Social justice and entrepreneurial identity: Models for the TPC classroom. In *Proceedings of the 38th ACM International Conference on Design of Communication* (pp. 1–8). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3380851.3416776

Bellini, R. (2023). Paying the price: When intimate partners use technology for financial harm. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (pp. 1–17). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3544548.3581101

Bellini, R., de Castro Leal, D., Dixon, H. A., Fox, S. E., & Strohmayer, A. (2022). "There is no justice, just us": Making mosaics of justice in social justice human–computer interaction. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (pp. 1–6). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3491101.3503698

Browne, P. S. (1976). Computer security: a survey. In *Proceedings of the June 7-10, 1976, National Computer Conference and Exposition (AFIPS '76)* (pp. 53–63). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/1499799.1499809

Campbell, N. (2004). Making sense of imbrication: popular technology and "inside-out" methodologies. In *Proceedings of the eighth conference on Participatory design: Artful integration: interweaving media, materials and practices - Volume 1 (PDC 04)* (pp. 65–73). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/1011870.1011878

Casey, D., Charvet, J., Miller, M., & Peretti, J. H. (1998). Virtual reality in education: Irish and American students meet on the virtual frontier. In *ACM SIGGRAPH 98 Conference abstracts and applications (SIGGRAPH '98)* (pp. 60). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/280953.281002

Chen, C., & Bergholm, N. J. (2020). Citizen technical communicators: A transnational social justice analysis of grassroots genres during the novel coronavirus outbreak. In *Proceedings of the 38th ACM International Conference on Design of Communication* (pp. 1–8). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3380851.3416778

Chordia, I. (2022). Leveraging transformative justice in organizing collective action towards community safety. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (pp. 1–4). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3491101.3503820

Chordia, I., Baltaxe-Admony, L. B., Boone, A., Sheehan, A., Dombrowski, L., Le Dantec, C. A., Ringland, K. E., & Smith, A. D. R. (2024). Social justice in HCI: A systematic literature review. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (pp. 1–33). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3613904.3642704

Collins, P. H. (2000) *Black Feminist Thought* (2nd). Routledge.

Corbett, E., & Loukissas, Y. (2019). Engaging gentrification as a social justice issue in HCI. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (pp. 1–16). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3290605.3300510

Costanza-Chock, S. (2020). *Design Justice: Community-Led Practices to Build the Worlds We Need*. The MIT Press. https://library.oapen.org/handle/20.500.12657/43542

Crenshaw, K. (2013). Demarginalizing the intersection of race and sex: A black feminist critique of antidiscrimination doctrine, feminist theory and antiracist politics. In *Feminist legal theories* (pp. 23–51). Routledge. https://api.taylorfrancis.com/content/chapters/edit/download?identifierName=doi&identifierValue=10.4324/9781315051536-2&type=chapterpdf

de Castro Leal, D., Strohmayer, A., & Krüger, M. (2021). On activism and academia: Reflecting together and sharing experiences among critical friends. In *Proceedings of the 2021 CHI Conference*

Downloaded from https://academic.oup.com/iwc/article/37/6/553/8099903 by Umeå University user on 20 October 2025

on Human Factors in Computing Systems (pp. 1–18). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3411764.3445263

Dimond, J. P., Dye, M., Larose, D., & Bruckman, A. S. (2013). Hollaback! The role of storytelling online in a social movement organization. In *Proceedings of the 2013 Conference on Computer Supported Cooperative Work* (pp. 477–490). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/2441776.2441831

DiSalvo, C., Sengers, P., & Brynjarsdóttir, H. (2010). Mapping the landscape of sustainable HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 1975–1984). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/1753326.1753625

Dombrowski, L., Harmon, E., & Fox, S. (2016a). Social justice-oriented interaction design: outlining key design strategies and commitments. In *Proceedings of the 2016 ACM Conference on Designing Interactive Systems* (pp. 656–671). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/2901790.2901861

Dourish, P. (2006). Implications for design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 541–550. https://doi.org/10.1145/1124772.1124855

Edenfield, A. C., Cheek, R., & Clem, S. (2021). Trans∗ vulnerability and digital research ethics: A Qubit ethical analysis of transparency activism. In *Proceedings of the 39th ACM International Conference on Design of Communication* (pp. 101–107). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3472714.3473628

Erete, S., Rankin, Y. A., & Thomas, J. O. (2021). I can't breathe: Reflections from Black women in CSCW and HCI. In *Proceedings of the ACM on Human-Computer Interaction*, **4**(CSCW3), 234:1–234:23. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3432933

Erete, S., Rankin, Y., & Thomas, J. (2023). A method to the madness: applying an intersectional analysis of structural oppression and power in HCI and design. *ACM Transactions on Computer-Human Interaction*, **30**(2), 24:1–24:45. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3507695

Evans, D. (2017). Keynote: Truth, social justice (and the American Way?). In *Proceedings of the 2017 Workshop on Women in Cyber Security* (pp. 3–4). Association for Computing Machinery, New York, NY, USA https://doi.org/10.1145/3139531.3139533

Ferreira, R., & Vardi, M. Y. (2021). Deep Tech Ethics: An Approach to Teaching Social Justice in Computer Science. In *Proceedings of the 52nd ACM Technical Symposium on Computer Science Education*, 1041–1047). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3408877.3432449

Flanagan, R. K. (2020). Principles of technical and public communication as a provisional framework for undergraduate students writing with and about indigenous peoples. In *Proceedings of the 38th ACM International Conference on Design of Communication*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3380851.3418617

Fonteles, J. H., Akpanoko, C., Wisniewski, P., & Biswas, G. (2024). *Promoting Equitable Learning Outcomes for Underserved Students in Open-Ended Learning Environments*. Association for Computing Machinery, New York, NY, USA https://stirlab.org/wp-content/uploads/Fonteles-et-al_idc24-4.pdf

Fox, S., Asad, M., Lo, K., Dimond, J. P., Dombrowski, L. S., & Bardzell, S. (2016). Exploring social justice, design, and HCI. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (pp. 3293–3300). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/2851581.2856465

Fox, S., Dimond, J., Irani, L., Hirsch, T., Muller, M., & Bardzell, S. (2017). Social justice and design: Power and oppression in collaborative systems. In *Companion of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing* (pp. 117–122). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3022198.3022201

Frankena, W. K. (1966). *Some beliefs about justice*. University of Kansas, Department of Philosophy. https://kuscholarworks.ku.edu/bitstream/handle/1808/12382/Some%20Beliefs%20About%20Justice-1966.pdf

Fraser, N. (2009). *Scales of Justice: Reimagining Political Space in a Globalizing World* (Vol. 31). Columbia University Press.

Ganguly, P. (2021). Tweeting reproductive justice (RJ) advocacy through #AbortionMeraHaq*. In *Proceedings of the 39th ACM International Conference on Design of Communication* (pp. 108–117). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3472714.3473630

Gui, X., Luo, Y., Ding, X., Abdullah, S., Dixon, E., & Bardzell, S. (2023). Designing for and reflecting upon resilience in health and wellbeing. In *Companion Publication of the 2023 ACM Designing Interactive Systems Conference* (pp. 135–137). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3563703.3591456

Gutstein, E. (2012). *Reading and writing the world with mathematics: Toward a pedagogy for social justice*. Routledge. New York. 2006. https://www.taylorfrancis.com/books/mono/10.4324/9780203112946/reading-writing-world-mathematics-eric-gutstein

Harrison, S., Tatar, D., & Sengers, P. (2007). The three paradigms of HCI. In *Alt. Chi. Session at the SIGCHI Conference on human factors in computing systems* (pp. 1-18). San Jose, California, USA.

Hooks, B. (2000). *Feminist Theory : From Margin to Center* (2. ed.). Pluto Press.

Jiménez, P., Pascual, J., & Mejía, A. (2019). Towards a pedagogical model of social justice in engineering education. In *Proceedings of the 8th International Conference on Software and Information Engineering* (pp. 55–59). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3328833.3328861

John, R. (1971). *A Theory of Justice*. The Belknap Press of Harvard University Press, Cambridge, Massachusetts.

Johnson, I. G., & Crivellaro, C. (2021). Opening research commissioning to civic participation: creating a community panel to review the social impact of HCI research proposals. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3411764.3445113

Jonas, R. M., De, A., & Cotter, K. (2024). "We happen to be different and different is not bad": Designing for intersectional fat-positive information-seeking. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3613904.3642599

Jones, L. C. (2021). Online advocacy work: "palatable" platforms and privilege in GUI features on Twitter and Instagram. In *Proceedings of the 39th ACM International Conference on Design of Communication* (pp. 157–164). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3472714.3473636

Jones, C. (2023). Representations and privacy: Methodological questions for social media data. In *Proceedings of the 41st ACM International Conference on Design of Communication* (pp. 96–102). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3615335.3623017

Jost, J. T., & Kay, A. C. (2010). Social justice: History, theory, and research. In *Handbook of social psychology, Vol.* **2**, 5th ed (pp.

Downloaded from https://academic.oup.com/iwc/article/37/6/553/8099903 by Umeå University user on 20 October 2025

1122–1165). John Wiley & Sons, Inc. https://doi.org/10.1002/9780470561119.socpsy002030

Kannabiran, G., Ahmed, A. A., Wood, M., Balaam, M., Tanenbaum, T. J., Bardzell, S., & Bardzell, J. (2018). Design for sexual well-being in HCI. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1–7). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3170427.3170639

Kariotis, T., & Mir, D. J. (2020). Fighting Back algocracy: The need for new participatory approaches to technology assessment. In *Proceedings of the 16th Participatory Design Conference 2020—Participation(s) Otherwise - Volume 2* (pp. 148–153). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3384772.3385151

Klassen, S. (2023). Black innovation: Creating and appropriating technology to survive and thrive. In *Companion Proceedings of the 2023 ACM International Conference on Supporting Group Work* (pp. 75–76). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3565967.3571762

Krawczyk, M. (2010). A glimpse through the veil of ignorance: equality of opportunity and support for redistribution. *J. Public Econ.*, **94**(1), 131–141. https://doi.org/10.1016/j.jpubeco.2009.10.003

Lan Fang, M. (2022). Future of AgeTech: transdisciplinary considerations for equity, intersectionality, sustainability, and social justice. In *Proceedings of the 15th International Conference on PErvasive Technologies Related to Assistive Environments* (pp. 536–541). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3529190.3534757

Lane, L. (2018). Iteration for impact: exploring design thinking & designing for social change in client projects. In *Proceedings of the 36th ACM International Conference on the Design of Communication*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3233756.3233952

Lee-Cultura, S., & Giannakos, M. (2020). Embodied interaction and spatial skills: a systematic review of empirical studies. *Interact. Comput.*, **32**(4), 331–366. https://doi.org/10.1093/iwcomp/iwaa023

Lin, K. (2022). CS education for the socially-just worlds we need: the case for justice-centered approaches to CS in higher education. In *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education—Volume 1* (pp. 265–271). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3478431.3499291

Lötter, HPP (Hennie PP), Poverty, ethics and justice.1st ed., Cardiff, University of Wales Press, 2011.

Marks, R., & Stanfill, M. (2023). Methodological solutions for the challenges of studying racist communication on social media. In *Proceedings of the 41st ACM International Conference on Design of Communication* (pp. 70–76). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3615335.3623013

Mayhew, E. J., & Patitsas, E. (2023). Critical pedagogy in practice in the computing classroom. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1* (pp. 1076–1082). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3545945.3569840

McMenemy, D. (2021). No longer "neutral among ends"—Liberal versus communitarian ethics in library and information science. In *Diversity, Divergence, Dialogue: 16th International Conference, IConference 2021, Beijing, China, March 17–31, 2021, Proceedings, Part II* (pp. 207–214. https://doi.org/10.1007/978-3-030-71305-8_16

Miller, D. (1999). *Principles of Social Justice*. Harvard University Press, JSTOR. https://doi.org/10.2307/j.ctv1pdrq04. Accessed 7 Mar. 2025.

Mowshowitz, A. (1984). Wealth and jobs in the fifth new world. In *Proceedings of the 1984 Annual Conference of the ACM on The fifth Generation Challenge (ACM '84)* (p. 261). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/800171.809645

Mudliar, P. (2020). Whither humane–computer interaction? Adult and child value conflicts in the biometric fingerprinting for food. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–12). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3313831.3376564

Nee, J., Macfarlane Smith, G., Sheares, A., & Rustagi, I. (2021). Advancing social justice through linguistic justice: Strategies for building equity fluent NLP technology. In *Proceedings of the 1st ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization* (pp. 1–9). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3465416.3483301

Oden Choi, J., Hammer, J., Royal, J., & Forlizzi, J. (2020). Moving for the movement: Applying viewpoints and composition techniques to the design of online social justice campaigns. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (pp. 75–86). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3357236.3395435

Ogbonnaya-Ogburu, I. F., Smith, A. D. R., To, A., & Toyama, K. (2020). Critical race theory for HCI. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–16). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3313831.3376392

Öhlund, L. (2021). The collateral damage of digitalization during covid. *Interactions*, **28**(4), 36–38. https://doi.org/10.1145/3466158

Öhlund, L. (2023). A social justice-oriented perspective on older adults technology use in HCI. In Q. Gao & J. Zhou (Eds.), *Human Aspects of IT for the Aged Population* (pp. 519–532). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-34917-1_35

Öhlund, L., & Almeida, T. (2023). Mapping the digital injustices of technology-facilitated sex trafficking. in J. Abdelnour Nocera, M. Kristín Lárusdóttir, H. Petrie, A. Piccinno, & M. Winckler (Eds.), *Human-Computer Interaction – INTERACT 2023* (pp. 523–527). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-42293-5_66

Ojedele-Adejumo, T. (2023a). Folkloric tradition as an epistemology of environmental justice. In *Proceedings of the 41st ACM International Conference on Design of Communication* (pp. 49–54). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3615335.3623010

Ojedele-Adejumo, T. (2023b). Mapping out the core constructs of social justice as heuristics in selected technical communication articles using the 4Rs model. In *Proceedings of the 41st ACM International Conference on Design of Communication* (pp. 276–277). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3615335.3623052

Opel, D. (2014). Social justice in technologies of prenatal care: Toward a user centered approach to technical communication in home pregnancy testing. In *Proceedings of the 32nd ACM International Conference on The Design of Communication CD-ROM* (pp. 1–8). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/2666216.2666223

Padiyath, A., Ashburn, K., & Ericson, B. (2024). Undergraduate student attitudes towards a social justice context in a programming project. *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (pp. 1000–1006). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3626252.3630955

Downloaded from https://academic.oup.com/iwc/article/37/6/553/8099903 by Umeå University user on 20 October 2025

Payton, J., Burge, J. D., & Denner, J. (2019). The reality of inclusion: The role of relationships, identity, and academic culture in inclusive and equitable practices for broadening participation in computing education. In *Proceedings of the 50th ACM Technical Symposium on Computer Science Education* (pp. 494–495). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3287324.3287337

Pendse, S. R., Sharma, A., Vashistha, A., De Choudhury, M., & Kumar, N. (2021). "Can I Not Be Suicidal on a Sunday?": Understanding technology-mediated pathways to mental health support. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3411764.3445410

Pritchard, H., Snodgrass, E., Morrison, R. R., Britton, L., & Moll, J. (2020). Burn, dream and reboot! Speculating backwards for the missing archive on non-coercive computing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, **683**. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3351095.3375697

Prost, S., Crivellaro, C., Haddon, A., & Comber, R. (2018). Food democracy in the making: Designing with local food networks. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1–14). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3173574.3173907

Rabaan, H. (2021). Exploring transformative justice principles to inform survivor-centered design for Muslim women in the United States. In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing* (pp. 291–294). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3462204.3481797

Rawls, J. (1971). *A theory of justice*. Belknap press, https://doi.org/10.4159/9780674042605.

Rose, E. J. (2016). Design as advocacy: using a human-centered approach to investigate the needs of vulnerable populations. *J. Tech. Writ. Commun.*, **46**(4), 427–445. https://doi.org/10.1177/0047281616653494

Rose, E. J., & Walton, R. (2015). Factors to actors: Implications of posthumanism for social justice work. In *Proceedings of the 33rd Annual International Conference on the Design of Communication* (pp. 1–10). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/2775441.2775464

Rose, E. J., Edenfield, A., Walton, R., Gonzales, L., McNair, A. S., Zhvotovska, T., Jones, N., Jones, N., Garcia de Mueller, G. I., & Moore, K. (2018). Social Justice in UX: Centering Marginalized Users. In *Proceedings of the 36th ACM International Conference on the Design of Communication (SIGDOC '18)*. *Article* **21**, 1–2. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3233756.3233931

Samuels, J. T., Mathew, A. P., Kommanivanh, C., Kwon, D., Gomez, L., Thunder, B. A., Velazquez, D., Martinez, M., & LaQueens, L. (2018). Art, human computer interaction, and shared experiences: A gun violence prevention intervention. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1–4). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3170427.3186526

de Santana, V. F., Quigley, L. T., Coates, J., & Graves, M. (2024). Computational notebook as a tool for supporting Black engineers nonprofit organizations in the context of workforce participation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. *Article* **512**, (pp. 1–9). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3613905.3637130

Sharma, V., Kumar, N., & Nardi, B. (2023). Post-growth human–computer interaction. *ACM Transactions on Computer-Human Interaction*, **31**(1), 9:1–9:37. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3624981

Sixsmith, J. A. (2022). Key ethical challenges in the AgeTech sector. In *Proceedings of the 15th International Conference on PErvasive Technologies Related to Assistive Environments* (pp. 532–535). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3529190.3534759

Smith, A. (2023). Designing feminist methodologies: Foregrounding gender, positionality, and justice in communication design research. In *Proceedings of the 41st ACM International Conference on Design of Communication* (pp. 39–48). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3615335.3623008

Steiger, M., Bharucha, T. J., Venkatagiri, S., Riedl, M. J., & Lease, M. (2021). The psychological well-being of content moderators: The emotional labor of commercial moderation and avenues for improving support. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3411764.3445092

Stevens, A. B. (2022). Including citizen's voices in economic growth: An exploration into the rhetoric of socio-economic change in Ukraine. In *Proceedings of the 40th ACM International Conference on Design of Communication* (pp. 183–185). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3513130.3559007

Strohmayer, A., Laing, M., & Comber, R. (2017). Technologies and social justice outcomes in sex work charities: Fighting stigma, saving lives. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (pp. 3352–3364). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3025453.3025615

Strohmayer, A., Clamen, J., & Laing, M. (2019). Technologies for social justice: Lessons from sex workers on the front lines. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, (pp. 1–14). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3290605.3300882

Sultana, S., Guimbretière, F., Sengers, P., & Dell, N. (2018). Design within a patriarchal society: Opportunities and challenges in designing for rural women in Bangladesh. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). Association for Computing Machinery. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3173574.3174110

Sultana, S., Pritha, S. T., Tasnim, R., Das, A., Akter, R., Shaid Hasan, S. M. Alam, R., Kabir, M. A., & Ahmed, S. I. (2022). 'ShishuShurokkha': A transformative justice approach for combating child sexual abuse in Bangladesh. In *International Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3491102.3517543

Swacha, K. Y. (2022). Social Justice and the Technical and Professional Communication Internship. In *Proceedings of the 40th ACM International Conference on Design of Communication* (pp. 136–139). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3513130.3558991

Tacheva, J., Namvarrad, S., & Almissalati, N. (2022). A higher purpose: towards a social justice informatics research framework. In *Information for a Better World: Shaping the Global Future: 17th International Conference, IConference 2022, Virtual Event, February 28–March 4, 2022, Proceedings, Part I* (pp. 265–271. https://doi.org/10.1007/978-3-030-96957-8_24

Downloaded from https://academic.oup.com/iwc/article/37/6/553/8099903 by Umeå University user on 20 October 2025

Tang, Y. (2020). Promoting user advocacy through design thinking in the age of automated writing. In *Proceedings of the 38th ACM International Conference on Design of Communication*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3380851.3416784

Tham, J. C. K. (2019). Feminist design thinking: A norm-creative approach to communication design. In *Proceedings of the 37th ACM International Conference on the Design of Communication*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3328020.3353919

Tricco, A. C., Lillie, E., Zarin, W., O'Brien, K., Colquhoun, H., Kastner, M., Levac, D., Ng, C., Sharpe, J. P., Wilson, K., Kenny, M., Warren, R., Wilson, C., Stelfox, H. T., & Straus, S. E. (2016). A scoping review on the conduct and reporting of scoping reviews. *BMC Med. Res. Methodol.*, **16**(1), 15. https://doi.org/10.1186/s12874-016-0116-4

Tseng, E., Okeke, F., Sterling, M., & Dell, N. (2020). "We can learn. Why not?": Designing technologies to engender equity for home health aides. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–14). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3313831.3376633

Tushar, A. K., Antara, I. J., Das, D., Chandra, P., Soron, T. R., Haque, M. M., Ahamed, S. I., & Ahmed, S. I. (2020). We need more power to stand up: designing to combat stigmatization of the caregivers of children with autism in Urban Bangladesh. In *Proceedings of the 2020 International Conference on Information and Communication Technologies and Development*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3392561.3394643

Van Wart, S. J., Vakil, S., & Parikh, T. S. (2014). Apps for social justice: Motivating computer science learning with design and real-world problem solving. In *Proceedings of the 2014 Conference on Innovation & Technology in Computer Science Education* (pp. 123–128). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/2591708.2591751

Walls, D. M. (2016). User experience in social justice contexts. In *Proceedings of the 34th ACM International Conference on the Design of Communication* (pp. 1–6). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/2987592.2987604

Walton, R., Moore, K., & Jones, N. (2019). *Technical Communication After the Social Justice Turn: Building Coalitions for Action*. Routledge. https://doi.org/10.4324/9780429198748

Westbrook, L. (2015). ASIS&T addressing missed information opportunities and victims' information agency in the fight against sexual human trafficking. *Proc. Assoc. Inf. Sci. Technol.*, **52**(1), 1–16. https://doi.org/10.1002/pra2.2015.145052010053

Wiberg, M., & Stolterman, E. (2021). Time and temporality in HCI research. *Interact. Comput.*, **33**(3), 250–270. https://doi.org/10.1093/iwc/iwab025

Wieczorek, C., Biggs, H., Jack, M. C., Forlano, L., & Bardzell, S. (2023). Collective healing to support design futures: Building community and exploring methods. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3544549.3573810

Young, I. M. (2020). Justice and the politics of difference. In *The new social theory reader* (pp. 261–269). Routledge. https://www.taylorfrancis.com/chapters/edit/10.4324/9781003060963–43/justice-politics-difference-iris-marion-young

Zhu, Y., Wen, R., & Williams, T. (2024). Robots for Social Justice (R4SJ): Toward a more equitable practice of human–robot interaction. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 850–859). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3610977.3634944

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